Solve and check division problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: ___________________

1. Read the model: A charity packs 9,360 tins equally into 36 crates. How many tins per crate? Copy or complete the first step.

2. 4,725 pupils travel in coaches holding 48. How many coaches are needed?

3. Calculate 7,488 / 26 and check by multiplication.

4. Miro says: Miro accepts a remainder that is larger than the divisor. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A charity packs 9,360 tins equally into 36 crates. How many tins per crate? Copy or complete the first step.
   Answer 260.

2. 4,725 pupils travel in coaches holding 48. How many coaches are needed?
   Answer 99 coaches.

3. Calculate 7,488 / 26 and check by multiplication.
   Answer 288 because 288 x 26 = 7,488.

4. Miro says: Miro accepts a remainder that is larger than the divisor. Is this correct? Explain.
   Answer A valid remainder must be smaller than the divisor.

Solve and check division problems

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. A charity packs 9,360 tins equally into 36 crates. How many tins per crate?

2. 4,725 pupils travel in coaches holding 48. How many coaches are needed?

3. Calculate 7,488 / 26 and check by multiplication.

4. Solve this independently and show every stage: A charity packs 9,360 tins equally into 36 crates. How many tins per crate?

5. Use a second method or representation for: 4,725 pupils travel in coaches holding 48. How many coaches are needed?

6. Check this result using an inverse, estimate or known fact: Calculate 7,488 / 26 and check by multiplication.

7. Create a division with divisor 28, quotient 135 and remainder 7.

8. Identify and correct this misconception: Miro accepts a remainder that is larger than the divisor.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Calculate 7,488 / 26 and check by multiplication.

Teacher answers and checking prompts

1. A charity packs 9,360 tins equally into 36 crates. How many tins per crate?
   Answer 260.

2. 4,725 pupils travel in coaches holding 48. How many coaches are needed?
   Answer 99 coaches.

3. Calculate 7,488 / 26 and check by multiplication.
   Answer 288 because 288 x 26 = 7,488.

4. Solve this independently and show every stage: A charity packs 9,360 tins equally into 36 crates. How many tins per crate?
   Answer 260.

5. Use a second method or representation for: 4,725 pupils travel in coaches holding 48. How many coaches are needed?
   Answer Expected result: 99 coaches. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Calculate 7,488 / 26 and check by multiplication.
   Answer Expected result: 288 because 288 x 26 = 7,488. A valid check must be shown.

7. Create a division with divisor 28, quotient 135 and remainder 7.
   Answer 3,787 / 28 = 135 remainder 7.

8. Identify and correct this misconception: Miro accepts a remainder that is larger than the divisor.
   Answer A valid remainder must be smaller than the divisor.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Calculate 7,488 / 26 and check by multiplication.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Solve and check division problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Create a division with divisor 28, quotient 135 and remainder 7.

2. Prove the answer to this question, rather than only calculating it: A charity packs 9,360 tins equally into 36 crates. How many tins per crate?

3. Solve this in two different ways and compare them: 4,725 pupils travel in coaches holding 48. How many coaches are needed?

4. Work backwards from the answer to reconstruct the question: 288 because 288 x 26 = 7,488.

5. Explain why this reasoning fails: Miro accepts a remainder that is larger than the divisor.

6. Create a new example that tests the same idea as: Calculate 7,488 / 26 and check by multiplication.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Create a division with divisor 28, quotient 135 and remainder 7.
Answer 3,787 / 28 = 135 remainder 7.

2. Prove the answer to this question, rather than only calculating it: A charity packs 9,360 tins equally into 36 crates. How many tins per crate?
Answer Expected result: 260. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: 4,725 pupils travel in coaches holding 48. How many coaches are needed?
Answer Expected result: 99 coaches. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 288 because 288 x 26 = 7,488.
Answer One valid reconstruction is: Calculate 7,488 / 26 and check by multiplication. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro accepts a remainder that is larger than the divisor.
Answer A valid remainder must be smaller than the divisor.

6. Create a new example that tests the same idea as: Calculate 7,488 / 26 and check by multiplication.
Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.